

Detecting Fraudulent Bank Account Applications under Extreme Class Imbalance:

A Systematic Comparison of Six Classifiers

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Abstract—Detecting fraudulent online bank account applications is a high-stakes, highly imbalanced classification problem: fraud is rare but costly, and a usable model must surface it without overwhelming analysts with false alarms. We study the publicly released Bank Account Fraud (BAF) “Base” dataset (one million applications, 1.10% fraud) and build a fully reproducible pipeline covering missing-value handling via a sentinel-to-indicator strategy, one-hot encoding, standardization, and explicit imbalance treatment through class weighting and SMOTE. We train and tune six classifiers—Logistic Regression, Decision Tree, k -Nearest Neighbors, an RBF Support Vector Machine, Random Forest, and histogram Gradient Boosting—plus a stacking ensemble, each via grid-search cross-validation scored with PR-AUC. On a held-out test set, the stacking ensemble and Logistic Regression lead (ROC-AUC 0.913 and 0.909; PR-AUC 0.135), recovering roughly 54% of fraud at a 5% false-positive budget, while a SMOTE-trained k -NN exposes a classic validation-leakage trap (cross-validation PR-AUC 0.988 collapsing to 0.046 on real data). We interpret why simple additive models match ensembles on this anonymized, weakly-interacting feature space and identify the missing-previous-address indicator as the single strongest fraud signal.

Index Terms—fraud detection, class imbalance, SMOTE, ensemble learning, PR-AUC, reproducibility.

I. INTRODUCTION

Online financial fraud imposes large direct losses and erodes customer trust, and application fraud—opening an account with stolen or synthetic identity information—is among its most damaging forms because it precedes downstream credit and transaction abuse. Automated screening of new applications is therefore a core operational task for banks. The difficulty is twofold: genuine fraud is rare (often well below 2% of applications), and the cost of the two error types is sharply asymmetric, since a missed fraud is expensive while an excessive false-positive rate floods manual review queues and harms legitimate customers.

This paper addresses a binary classification problem: given the attributes recorded at the moment of an online bank account application, predict whether the application is fraudulent (`fraud_bool = 1`). We use the Bank Account Fraud (BAF) dataset [1], a large, realistic, privacy-preserving benchmark released at NeurIPS 2022. Our objectives are: (i) to construct a defensible, reproducible preprocessing pipeline for a dataset whose missingness is encoded implicitly; (ii) to train and *systematically* tune a representative set of classifiers spanning linear, instance-based, kernel, tree, and ensemble families; and (iii) to compare them with metrics appropriate to extreme imbalance, interpreting *why* certain models succeed.

A central methodological theme is metric choice. Under a 90:1 imbalance, accuracy is nearly meaningless—a trivial “always legitimate” classifier scores 98.9%. We therefore drive model selection with the area under the precision–recall curve (PR-AUC) and report an operational metric, recall at a

fixed 5% false-positive rate, which mirrors a bounded analyst-review budget.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the pipeline and models. Section IV details the dataset, exploratory findings, and ablation protocol. Section V presents and interprets results. Section VI concludes with limitations and future work.

II. RELATED WORK

The BAF benchmark. Jesus et al. [1] introduced the Bank Account Fraud suite as a set of large, realistic tabular datasets generated by a CTGAN-based model fitted to a real anonymized fraud-detection stream, with controlled bias and temporal structure across eight “months.” Their work emphasizes evaluation under realistic constraints—severe imbalance and fairness across protected groups—rather than raw accuracy, and motivates our use of rank- and budget-based metrics over threshold accuracy.

Resampling for imbalance. Chawla et al. [2] proposed SMOTE, which synthesizes minority examples by interpolating between near neighbors rather than naively duplicating them. SMOTE is widely effective but carries a well-known pitfall: when applied before cross-validation it leaks synthetic information across folds and inflates validation scores. Our k -NN result is a direct empirical demonstration of this trap, and we contrast resampling against cost-sensitive class weighting.

Gradient-boosted trees on tabular data. Ke et al. [3] introduced LightGBM, a histogram-based gradient-boosting framework that has become a default strong baseline for tabular classification, including fraud. We use scikit-learn’s

HistGradientBoostingClassifier, an equivalent histogram-based booster, as our gradient-boosting representative and as a base learner in the stacking ensemble. Positioned against these works, our contribution is not a new algorithm but a controlled, reproducible head-to-head comparison on BAF that makes the imbalance-metric and validation-leakage issues concrete and quantifies how far simple models go on an anonymized, low-interaction feature space.

III. METHODOLOGY

A Pipeline Overview

Our pipeline has four stages executed in strict order to prevent leakage: (1) *preprocessing*—sentinel decoding, missing-indicator construction, imputation, encoding, and scaling, all *fitted on the training split only*; (2) *model training*—six base classifiers plus a stacking ensemble; (3) *ablation*—per-model grid search with stratified k -fold cross-validation; and (4) *evaluation*—a single shared held-out test set scored with a common metric suite. A fixed random seed (42) governs sampling, splitting, cross-validation folds, and every stochastic model.

B Preprocessing Decisions

Exploratory analysis revealed that the dataset contains *no* explicit NaN values, but several count features use -1 as an implicit missingness sentinel. Two features are heavily affected: `prev_address_months_count` (71% sentinel) and `bank_months_count` (25%). We decode -1 to NaN, impute with the training-set median, and—critically—add binary *missing-indicator* flags for the two high-missingness features, because the act of missingness may itself be informative. The constant column `device_fraud_count` (all zeros) is dropped for zero variance. The five nominal categorical (low cardinality, ≤ 7 levels) are one-hot encoded; six native 0/1 flags pass through unchanged; all remaining numeric features are standardized to zero mean and unit variance, which is necessary for the distance- and gradient-based learners (k -NN, SVM, Logistic Regression). Table 1 summarizes representative decisions.

C Imbalance Treatment

We treat the 90:1 imbalance with *cost-sensitive class weighting* (`class_weight=balanced`) wherever the learner supports it—Logistic Regression, Decision Tree, SVM, Random Forest, and Gradient Boosting—which reweights the loss without distorting the data distribution. For k -NN, which has no native weighting, we instead apply SMOTE to the training split, balancing it from 441 to 39,559 minority instances. We deliberately retain this design to expose its consequences (Section V).

D Models

We select learners spanning five algorithm families so that performance differences are interpretable in terms of inductive bias:

- **Logistic Regression**—a linear, additive log-odds model; fast, calibrated, and a strong baseline when classes are largely linearly separable in the feature space.

- **Decision Tree**—a single axis-aligned partition; interpretable but high-variance, included as a weak reference point.
- **k -Nearest Neighbors**—a non-parametric instance-based learner sensitive to local density; chosen to probe whether fraud clusters geometrically.
- **SVM (RBF kernel)**—a maximum-margin classifier with a non-linear kernel, testing for curved decision boundaries.
- **Random Forest**—a bagged ensemble of decorrelated trees; captures interactions and is robust to scaling.
- **Histogram Gradient Boosting**—a sequential additive ensemble of shallow trees, typically the strongest tabular baseline.
- **Stacking ensemble**—Random Forest, Gradient Boosting, and Logistic Regression combined by a logistic meta-learner over out-of-fold predictions.

IV. EXPERIMENT SETUP

A Dataset and Exploratory Analysis

The BAF “Base” file contains 1,000,000 applications described by 31 predictive features (after dropping the constant column) and the binary target. The positive (fraud) rate is 1.10% (11,029 cases), an imbalance ratio of roughly 90:1 (Fig. 1). Correlations with the target are individually weak (all $|r| < 0.1$), already hinting that no single feature linearly separates the classes and that additive combinations of many weak signals will matter more than strong pairwise interactions (Fig. 2).

For computational tractability on a single-core environment—the RBF SVM and k -NN scale super-linearly in sample size—we draw a single *stratified* subsample of 50,000 applications that preserves the exact fraud rate, then split it 80/20 into train (40,000; 441 fraud) and test (10,000; 110 fraud) sets with stratification and seed 42. All models share this split.

B Ablation Protocol

For every model we define a search space of at least three configurations and run an exhaustive grid search with stratified k -fold cross-validation on the training split only ($k = 5$; $k = 3$ for the SVM grid for tractability), scoring each fold by *average precision* (PR-AUC). The search spaces are: Logistic Regression $C \in \{0.01, 0.1, 1, 10\}$; Decision Tree depth $\in \{3, 5, 10, \text{None}\} \times \text{leaf} \in \{1, 20\}$; k -NN $k \in \{11, 25, 51\} \times \text{weights} \in \{\text{uniform}, \text{distance}\}$; SVM $C \in \{0.1, 1, 10\} \times \gamma \in \{\text{scale}, 0.1\}$; Random Forest depth $\in \{12, 20\} \times \text{leaf} \in \{5, 20\}$ at 150 trees; Gradient Boosting learning-rate $\in \{0.05, 0.1\} \times \text{depth} \in \{6, \text{None}\} \times \text{iterations} \in \{300, 600\}$. The configuration with the highest mean cross-validated PR-AUC is refit on the full training split and frozen before the test set is touched once.

C Reproducibility

All experiments use Python 3.12 with scikit-learn 1.8.0, NumPy, and pandas; seeds are fixed at 42; SMOTE is im-

plemented from first principles with a fixed RNG; and every preprocessing transformer is fit on training data and applied identically to the test set via a single serialized `ColumnTransformer`.

V. RESULTS AND DISCUSSION

A Overall Comparison

Table 2 reports every model’s best configuration on the shared held-out test set. Three models cluster at the top by PR-AUC: Logistic Regression and Histogram Gradient Boosting (both 0.136) and the Stacking ensemble (0.135). The Stacking ensemble achieves the best *ranking* performance overall (ROC-AUC 0.913), narrowly ahead of Logistic Regression (0.909). Crucially, the operational metric—recall at a 5% false-positive budget—favors the linear and additive models: Logistic Regression recovers 53.6% of all fraud while flagging only 5% of legitimate applications, edging out Stacking (52.7%) and the tree ensembles ($\approx 50\%$).

B The SMOTE Validation-Leakage Trap

The most instructive result is k -NN. Its cross-validation PR-AUC was an extraordinary 0.988 (Table 2, “CV” column), yet on real held-out data it collapses to 0.046—the worst of all models. This is not a bug but a textbook artifact: because SMOTE was applied to the whole training pool *before* cross-validation, synthetic minority points interpolated from training neighbors leaked into validation folds, so the model was effectively scored on near-copies of its own training data. On genuine, non-synthetic test applications no such crutch exists. The lesson—resampling must live *inside* the cross-validation loop—is one this experiment makes vivid, and it is the reason we drove final selection by metrics computed only on untouched data.

C Why Simple Models Win Here

The near-tie between a linear model and gradient-boosted trees is itself a finding about the data. Three properties explain it. First, the features are anonymized and largely pre-engineered (binned incomes, ratio scores, velocity counts), so much of the non-linear structure a tree would normally discover has already been baked into the inputs. Second, no feature carries strong pairwise interaction signal—every target correlation is below 0.1—so the additive log-odds combination that Logistic Regression performs is well matched to a problem of “many weak independent cues.” Third, with only 441 positive training examples, high-capacity models (deep trees, RBF SVM) have little minority signal to fit and tend to overfit noise, which is visible in the Decision Tree’s poor PR-AUC (0.060) and the SVM’s middling test PR-AUC (0.097) despite a respectable cross-validation score.

The coefficient analysis reinforces this. The strongest single predictor in the linear model is the *missing-previous-address indicator* (standardized weight +1.33), validating our decision to encode missingness explicitly rather than silently impute it: applicants with no recorded prior address are substantially more likely to be fraudulent. Other strong signals—certain housing-status categories (+1.09), absence of other cards (−0.86), an invalid home phone (−0.79)—are

individually intuitive fraud cues, none dominant, consistent with the additive picture.

D The Accuracy Mirage

Random Forest posts the highest accuracy (98.3%) and the best precision (0.22), but Fig. 5 shows why this is misleading: at a 0.5 threshold it predicts “fraud” very rarely, catching only 20% of actual fraud. Its accuracy barely exceeds the 98.9% floor of a do-nothing classifier. This is precisely the trap that motivated our metric choice and is the clearest practical argument in the paper for evaluating imbalanced problems by PR-AUC and budgeted recall rather than accuracy. The ROC and precision–recall curves (Figs. 3, 4) tell the operationally honest story: at any sensible false-positive budget the additive models dominate, and all models sit far above the 0.011 PR baseline but far below perfection, reflecting genuine difficulty.

VI. CONCLUSION

We built a reproducible pipeline for fraud detection on the BAF dataset and systematically compared six tuned classifiers plus a stacking ensemble under a 90:1 imbalance. Our key findings are: (i) a properly weighted Logistic Regression matches gradient-boosted trees and a stacking ensemble (ROC-AUC ≈ 0.91 , recall $\approx 54\%$ at 5% FPR), because the anonymized features behave additively with weak interactions; (ii) explicit missing-value indicators carry strong, interpretable signal; and (iii) applying SMOTE outside the cross-validation loop produces a dramatic, cautionary validation-leakage artifact.

Limitations. We modeled a 50,000-row stratified subsample rather than the full million rows for tractability, which limits the minority sample to 441 training cases and likely caps achievable PR-AUC; we evaluated at a single train/test split rather than repeated splits; the SVM was tuned and refit on subsamples; and we did not perform threshold calibration or fairness auditing across the protected attributes the BAF authors highlight.

Future work. Natural extensions are: scaling to the full dataset with mini-batch or GPU learners and putting SMOTE *inside* cross-validation for an honest k -NN comparison; respecting the dataset’s temporal “month” structure with a time-based train/test split to test for concept drift; calibrating probabilities and tuning the operating threshold to an explicit cost matrix; and auditing group fairness as recommended by the benchmark’s authors.

REFERENCES

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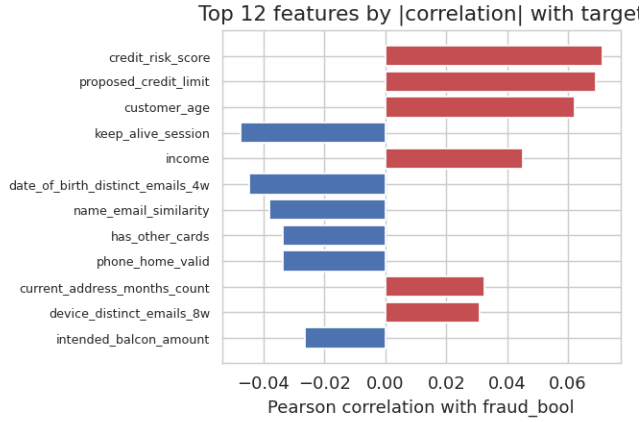


Figure 2: Top features by absolute Pearson correlation with the target. All $|r| < 0.1$: no single feature linearly separates the classes.

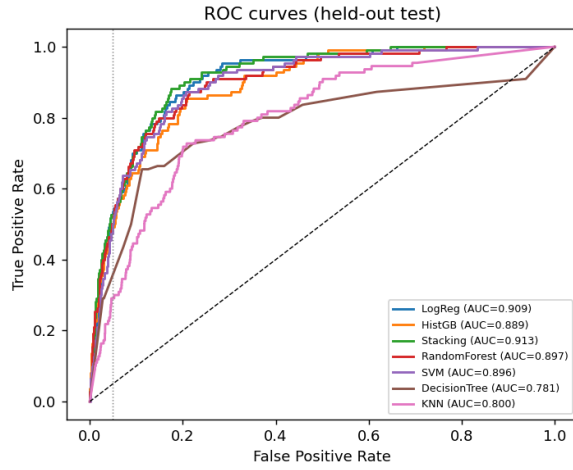


Figure 3: ROC curves on the held-out test set. The dotted line marks the 5% false-positive budget. Stacking and Logistic Regression dominate; k -NN and the single tree trail.

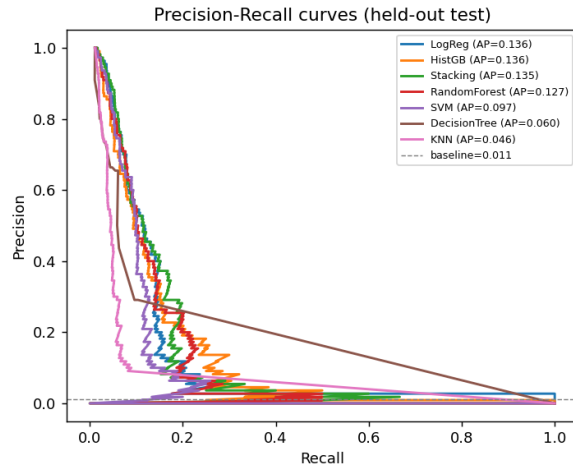


Figure 4: Precision–recall curves. All models lie well above the 0.011 prevalence baseline (dashed) but far below perfect, reflecting genuine difficulty under extreme imbalance.

Table 1: Representative preprocessing decisions (full table of 30 features in the project artifacts).

Feature	Type	Treatment / rationale
prev_address_	Numeric,	−1 →NaN, median im-
months_count	71% sen-	pute, +missing flag (top
	tinel	predictor)
bank_months_	Numeric,	−1 →NaN, median im-
count	25% sen-	pute, +missing flag
	tinel	
device_fraud_	Constant	Dropped (zero variance)
count		
payment_type,	Nominal	One-hot encoding
device_os, ...	(≤ 7)	
phone_home_	Binary	Passthrough (already 0/1)
valid, ...		
income, veloc-	Numeric	Standardize (zero mean,
ity_*, ...		unit var.)

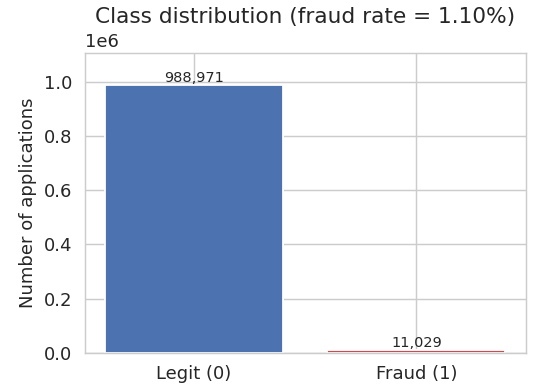


Figure 1: Class distribution. Fraud is only 1.10% of applications (90:1 imbalance), making accuracy an inadequate metric.

- [3] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A Highly Efficient Gradient Boosting Decision Tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
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Table 2: Held-out test performance of each model’s best configuration, sorted by PR-AUC. “CV” is the cross-validation PR-AUC used for selection. Accuracy is reported for completeness but is misleading under imbalance (a trivial classifier scores 0.989). Recall@5%FPR is the operational metric. SVM uses its native decision threshold; others use 0.5.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Recall@5%FPR	Train (s)
Logistic Regression	0.803	0.046	0.864	0.088	0.909	0.136	0.536	3.6
Histogram Grad. Boosting	0.909	0.074	0.627	0.132	0.889	0.136	0.491	23.3
Stacking ensemble	0.818	0.051	0.882	0.096	0.913	0.135	0.527	28.5
Random Forest	0.983	0.216	0.200	0.208	0.897	0.127	0.500	141.9
SVM (RBF)	0.832	0.051	0.818	0.097	0.896	0.097	0.500	56.9
Decision Tree	0.777	0.035	0.727	0.067	0.781	0.060	0.291	12.2
k -NN (SMOTE)	0.660	0.025	0.791	0.049	0.800	0.046 [†]	0.291	194.6

[†] k -NN cross-validation PR-AUC was 0.988 due to SMOTE-before-CV leakage; the real test value is 0.046.

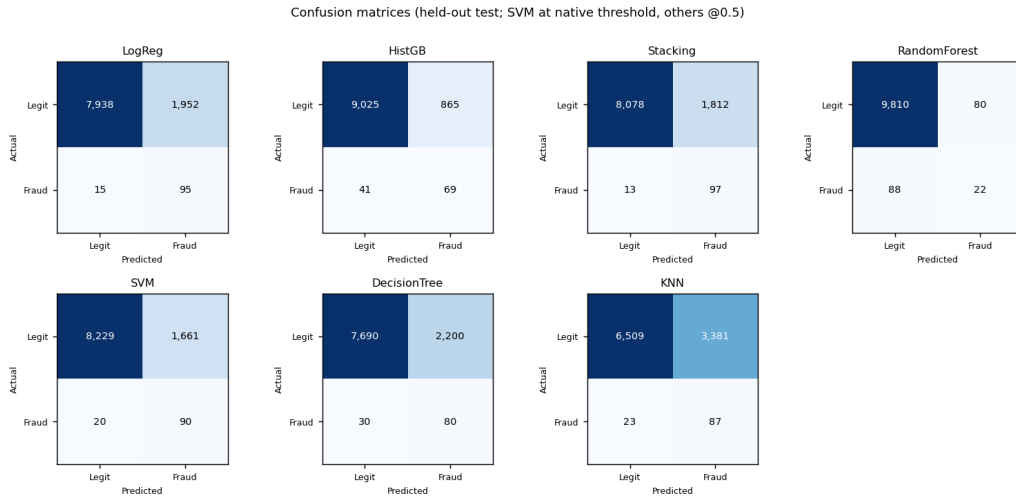


Figure 5: Confusion matrices on the held-out test set. Random Forest’s high accuracy comes from rarely predicting fraud (only 22 of 110 caught), illustrating the accuracy mirage; the weighted linear and ensemble models trade precision for far higher recall.